MIL-C-55040B(MU)
30 December 1969
SUPERSEDING
MIL-C-55040A(MU)
12 February 1963

(PROPOSED)
MILITARY SPECIFICATION

CABLE, SPECIAL PURPOSE, ELECTRICAL (RETRACTILE)

1. SCOPE

1.1 This specification covers one type, single conductor, shielded, retractile electrical cable.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-W-343	Wire, Electrical and Nonelectrical, Copper (Uninsulated)
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard
PPP-T-76	Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant

Military

MIL-P-116	Preservation, Methods of

STANDARDS

Federal

Fed. Test Method Std. No. 228	Cable and Wire, Insulated; Method of Testing

FSC 6145

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Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CONSOLIDATED CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, One Park Avenue at 33rd Street, New York, N.Y. 10016.)

AMERICAN TRUCKING ASSOCIATIONS, INC.
National Motor Freight Classification No. A-3 and No. 13

(Application for copies of the above publications should be addressed to the American Trucking Associations, Inc., Traffic Dept., 1424 16th Street, N.W., Washington, D.C. 20009.)

3. REQUIREMENTS

3.1 Conductor.- The conductor shall consist of 41 strands AWG No. 40 tin coated soft copper and shall conform to QQ-W-343 (Type B, Class Q) except that the lay shall be right hand and 5/8 inch maximum.

3.2 Insulation.- The stranded conductor shall be insulated with a rubber compound conforming to the following physical requirements, before and after aging, in accordance with Table II.

		Original	Aged
Tensile strength	(minimum) pounds per square inch (psi)	600	450
Elongation	(minimum) percent	250	175
Set	(maximum) inch	1/2	--

3.2.1 Color.- The color of the insulation applied over the conductor shall be white.

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3.2.2 Dimensions. - The overall diameter of the insulated conductor shall be 0.060 ± .003 inches nominal and the wall thickness at any point shall not be less than 0.010 inch.

3.3 Shield. - A shield shall be applied over the primary insulation and shall consist of 31 minimum strands AWG No. 36 tin coated soft copper layed parallel, helically served around the insulation, 33 turns minimum per foot and the individual strands shall conform to QQ-W-343 (Type S).

3.4 Jacket. - The insulation and shield shall be jacketed with a smooth compound of terpolymer of ethylene, propylene and diene conforming to the following physical requirements before and after aging in accordance with Table II.

	Original	Aged
Tensile strength psi (minimum)	1400	1300
Elongation percent (minimum)	350	300

3.4.1 Color. - The color of the jacket shall be black.

3.4.2 Dimensions. - The overall diameter of the jacket shall be 0.160 ± 0.010 inch and the wall thickness shall not be less than 0.025 inch.

3.5 Construction. - The jacketed electrical cable shall be helical in form with adjacent turns contiguous to form a helical cable having a straight length of cable on each end tangent to the helical coil. The finished electrical cable shall have a left lay in its helical section. The finished cable shall have a smooth and uniform jacket. The helix diameter shall be 9/16 inch ± 1/16 inch. The length of the helical portions, and the length of the straight ends shall be as specified in the detailed drawing or military standard sheet.

3.6 Extensile and retractile characteristics.

3.6.1 Conditioning. - The electrical cable shall be capable of returning to its original contiguous position after extension of the coiled portion of the cable to 5 times its retracted length for 6 cycles.

3.6.2 Life cycle. - The cable shall have a life cycle of 5000 cycles when extended to 4 times its retracted length at a rate not less than 18 cycles per minute (cpm) and shall be capable of returning to within 20 percent of its original retracted length within one hour after completion of the cycling. The change in resistance of the conductor at any time during the life cycle of the cable shall not exceed 10 percent of its original resistance.

3.6.3 Extended length. - The finished cable shall have a minimum extended length of 2-1/2 times its retracted length under a load of 6 ounces.

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3.6.4 Set.- After being suspended vertically to 5 times its retracted length and immediately released, the cable shall return to within 20 percent of its original vertical length within 5 minutes.

3.6.5 Static load.- Under vertical suspension due to static loading resulting in an extension of 2-1/2 times its original length for a period of 48 hours, the cable after being freed from restraint shall return to not more than 115 percent of its original retracted length within 30 minutes.

3.6.6 Retraction at low temperature.- During exposure, after conditioning for 20 hours at $-65^{\circ} \pm 2^{\circ}F$, the cable shall be capable of being extended to 4 times its retracted length without fracture of the conductor, conductor insulation or jacket. When released from its extended position, the cable shall retract to not more than 250 percent of its original length within 30 seconds.

3.7 Electrical.

3.7.1 Voltage withstand.- The electrical cable shall withstand for at least 5 seconds a direct potential of 500 volts (350 volts rms a.c.) without showing any signs of dielectric breakdown.

3.7.2 Insulation resistance.- The minimum value for insulation resistance shall be 1000 megohms when a direct current potential of 500 volts is applied for 1 minute.

3.8 Workmanship.- Cable furnished under this specification shall meet all the requirements of this specification and any referenced document. Only such dirt or foreign substances which would impair proper functioning of the cable need be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General provisions.

4.2.1 Definitions.- Definitions of inspection terms shall be as listed in MIL-STD-109.

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4.2.2 Lot formation.- Unless otherwise specified by the contracting officer, or his representative, inspection by lot sizes and lot formation shall be in accordance with MIL-STD-105.

4.2.3 Examination.- The classification of defects in Tables I thru IV, shall constitute the minimum inspection requirements to be performed prior to acceptance.

4.2.4 Inspection conditions.- Unless otherwise specified all inspection and testing shall be performed at room ambient temperature, pressure and humidity.

4.3 Acceptance inspection.- Acceptance inspection shall consist of Group A, B and C inspection.

4.3.1 Group A inspection.- Group A inspection shall consist of examinations and tests specified in Table I. Sampling shall be in accordance with MIL-STD-105. AQL shall be 1.0 percent defective for major defects and 4.0 percent for minor defects.

TABLE I - GROUP A INSPECTION

Class	Inspection	Requirement	Test Procedure
Major			
101.	Voltage withstand	3.7.1	4.8.1
102.	Insulation resistance	3.7.2	4.8.2
Minor			
201.	Dimensions	3.2; 3.4.2	4.5
202.	Color	3.2.1; 3.4.1	4.5
203.	Construction	3.5	4.5

4.3.2 Group B inspection.- Group B inspection shall consist of the examinations and tests specified in Table II. Sampling for Group B inspection shall be in accordance with inspection level S-3, Table I, of MIL-STD-105. Acceptance or rejection of the lot shall be in accordance with AQL 4.0 of Table II-A of MIL-STD-105.

TABLE II - GROUP B INSPECTION

Inspection	Requirement Paragraph	Test Paragraph
Physical tests	3.2; 3.4	4.6
Conditioning	3.6.1	4.7.1.1
Extended length	3.6.3	4.7.1.3
Set	3.6.4	4.7.1.4
Static load	3.6.5	4.7.1.5
Retraction at low temperature	3.6.6	4.7.1.6

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4.3.3 Group C inspection.- Group C inspection shall consist of the test specified in Table III. One sample per five hundred units shall be selected for Group C inspection.

TABLE III - GROUP C INSPECTION

Inspection	Requirement Paragraph	Test Paragraph
Life cycle	3.6.2	4.7.1.2

4.3.3.1 Noncompliance.- If a sample unit fails Group C inspection, acceptance inspection shall be discontinued until the cause for failure has been determined and corrected.

4.4 Material.- The contractor shall supply the Government with certification that materials used in the conductor and shield shall be as specified in 3.1 and 3.3.

4.5 Visual and dimensional inspection.- Cables shall be inspected for conformance with the applicable paragraph of Table I for dimensions, color and construction.

4.6 Physical tests.

4.6.1 Tensile strength.- The insulation and jacket shall conform to the minimum values of 3.2 or 3.4, as applicable, when tested in accordance with Method 3021 of Fed. Test Method Std. No. 228.

4.6.2 Elongation.- The insulation and jacket shall conform to the minimum values of 3.2 or 3.4, as applicable, when tested in accordance with Method 3031 of Fed. Test Method Std. No. 228.

4.6.3 Set.- The insulation shall not exceed the maximum values of 3.2 when tested in accordance with Method 3161 of Fed. Test Method Std. No. 228 when the following exceptions are followed:

(a) Bench marks shall be 2 inches apart.
(b) The test specimen shall be stretched so that the distance between bench marks is 6 inches; they shall be released within 5 seconds.
(c) Set shall be determined 1 minute after beginning of release.

4.6.4 Accelerated aging.- The insulation and jacket shall conform to the minimum aged values of 3.2 or 3.4, as applicable, when tested in accordance with Method 4011 of Fed. Test Method Std. No. 228 except that the aging period shall be 96 hours $\pm$ 1/2 hour.

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4.7 Test procedures.

4.7.1 Mechanical tests.

4.7.1.1 Conditioning.- Prior to meeting any extensile and retractile tests, the cable shall first be conditioned by extending the coiled portion to 5 times its retracted length 6 times and allowing it to retract freely each time. The cable shall then conform to 3.6.1 when laid on a horizontal surface.

4.7.1.2 Life cycle.- The resistance of the conductor shall be measured and recorded. The cable shall then be clamped in a suitable testing machine and extended to 4 times its retracted length for 5000 cycles at a rate of not less than 18 cycles per minute. The stretching and resistance values shall be in accordance with 3.6.2.

4.7.1.3 Extended length.- The finished cable shall be positioned vertically and a load of 6 ounces applied to the end. Measurement under suspension shall conform to 3.6.3.

4.7.1.4 Set.- The cable shall be suspended vertically and extended to 5 times its retracted length and immediately released and inspected for compliance with 3.6.4.

4.7.1.5 Static load.- The finished cable shall be extended between two adjustable clamps fastened to a vertical support so that the coiled portion of the cable is stretched to 250 percent of its original retracted length. After 48 hours, the cable shall be released from its extended position and freed from restraint by tapping on a horizontal surface and laid in a relaxed position. 30 minutes after being placed at rest on the horizontal surface, the cable shall be inspected for conformance with 3.6.5.

4.7.1.6 Retraction at low temperature.- The finished electrical cable shall be placed in a temperature controlled chamber for not less than 20 hours at a temperature of $-65^{\circ} \pm 2^{\circ}F$. At the end of the period and without removing from the chamber the cable shall be extended to 4 times its retracted length and then released on a horizontal surface. While still at the $-65^{\circ}F$ temperature, the cable shall be free from restraint by tapping on the surface and inspected for conformance with 3.6.6.

4.8 Electrical tests.

4.8.1 Voltage withstand.- The voltage withstand test shall be conducted in accordance with Method 6111 of Fed. Test Method Std. No. 228 with 500 volts d.c. being applied between the conductor and the shield for a minimum of 5 seconds to determine compliance with 3.7.1.

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4.8.2 Insulation resistance.- The insulation resistance test shall be conducted in accordance with Method 6031 of Fed. Test Method Std. No. 228 using a direct current potential of 500 volts to determine compliance with 3.7.2.

4.9 Inspection of preparation for delivery.

4.9.1 Inspection.- Contractor inspection shall ascertain prior to packaging and packing that the cables have been inspected and tested in accordance with this specification.

4.9.2 Sampling.- Sampling, including lot size, shall be in accordance with MIL-STD-105. The AQL shall be 1.5 percent for major defects and 2.5 percent for minor defects.

4.9.3 Classification of defects.- Major and minor defects for the inspection of levels A and B shall be as defined in Table IV.

TABLE IV - CLASSIFICATION OF DEFECTS FOR LEVEL A & B

Inspection	Req. Par.	Major	Minor
Improper cleaning	5.1.1.1	X	
Improper packaging method applied	5.1.1.2; 5.2.2	X	
Improper box	5.1.1.2	X	
Improper material applied to closure of container	5.1.1.3		X
Gross weight of intermediate container exceeded	5.1.1.3		X
Improper intermediate container	5.1.1.3	X	
Improper closure of box	5.2.1; 5.2.2		X
Improper class, type box	5.2.1; 5.2.2	X	
Gross weight of shipping container exceeded	5.2.1; 5.2.2		X
Strapping not in accordance with applicable box specification	5.2.1	X	
Gross weight not within limitations of applicable box specification	5.2.1; 5.2.2	X	
Improper markings	5.3	X	

4.10 Level C preservation, packaging and packing.- Due to the variance of preservation, packaging and packing that would conform to this level of packaging and packing the contractor must submit his intended packaging and packing methods and inspection procedures to the procuring activity for approval.

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4.11 Weight.- Inspection for weight shall be performed utilizing test equipment capable of determining conformance to the gross weight requirements of the specification.

4.12 Disposition of rejected lots.- All rejected lots shall be returned to the contractor for screening of the defective characteristics. After corrections have been made, the lot shall be resubmitted for inspection as outlined in MIL-STD-105.

4.13 Inspection tests.- Each unit package under test shall be visually inspected in accordance with MIL-P-116 for conformance with 5.1.1.1, 5.1.1.2 and 5.1.1.3.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A.

5.1.1.1 Cleaning.- Each cable shall be cleaned by process C-1 of MIL-P-116.

5.1.1.2 Unit packaging.- Each cable shall be unit packaged Method III of MIL-P-116 and further packaged as specified in 5.1.1.3.

5.1.1.3 Intermediate packaging.- Identical cables (cables of the same stock number) packaged as specified in 5.1.1.2 shall be packaged in a close-fitting intermediate container conforming to PPP-B-636, W5c. Closure of the container shall be with tape conforming to PPP-T-76 as specified in the appendix of PPP-B-636. The gross weight shall not exceed approximately 20 pounds.

5.1.2 Level C.- Cleaning and packaging shall be in accordance with manufacturers commercial practice. The cables shall be protected to prevent deterioration during shipment and subsequent interval preliminary to use.

5.2 Packing.

5.2.1 Level A.- The cables packaged as specified in 5.1.1.3 shall be packed in a wirebound wood box, wood-cleated fiberboard box, wood-cleated plywood box, nailed wood box or a fiberboard box conforming to the requirements of PPP-B-585 class 3, PPP-B-591 (Overseas type), PPP-B-601 style A or B, type I or II, class 2, PPP-B-621 class 2 and PPP-B-636 class 2 or 3, respectively. Closure shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds. The gross weight of fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

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5.2.2 Level B.- The cables packaged as specified in 5.1.1.3 shall be packed in a wirebound wood box, wood-cleated fiberboard box, wood-cleated plywood box, nailed wood box or a fiberboard box conforming to the requirements of PPP-B-585 class 1, PPP-B-591 (Domestic type), PPP-B-601 (Domestic type), PPP-B-621 class 1 and PPP-B-636 class 1, respectively. Closure shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds. The gross weight of fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.2.3 Level C.- The cables packaged as specified in 5.1.2 shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Uniform Freight Classification Rules or Motor Freight Regulations, whichever is applicable.

5.3 Marking.- Unit packages, intermediate packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use.- The cables procured under this specification will be used for wiring instrument lights on fire control equipment and selected general applications. In use the cable will be subjected to rough handling and extreme weather conditions. Material used in the fabrication of these cables is not intended for extended exposure in oil.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Title, number, and date of drawing or military standard sheet.
(c) Selection of applicable levels of packaging and packing.

Custodian:
Army - MU

Review Activity
Army - EL

Preparing activity:
Army - MU

Project No. 6145-A007

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